Accuracy of Free Energies of Hydration Using CM1 and CM3 Atomic Charges

MARINA UDIER–BLAGOVIĆ, PATRICIA MORALES DE TIRADO, SHOSHANNAH A. PEARLMAN, WILLIAM L. JORGENSEN
Department of Chemistry, Yale University, P.O. Box 208107, New Haven, Connecticut 06520-8107

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Abstract: Absolute free energies of hydration (ΔG_{hyd}) have been computed for 25 diverse organic molecules using partial atomic charges derived from AM1 and PM3 wave functions via the CM1 and CM3 procedures of Cramer, Truhlar, and coworkers. Comparisons are made with results using charges fit to the electrostatic potential surface (EPS) from *ab initio* 6-31G* wave functions and from the OPLS-AA force field. OPLS Lennard–Jones parameters for the organic molecules were used together with the TIP4P water model in Monte Carlo simulations with free energy perturbation theory. Absolute free energies of hydration were computed for OPLS united-atom and all-atom methane by annihilating the solutes in water and in the gas phase, and absolute ΔG_{hyd} values for all other molecules were computed via transformation to one of these references. Optimal charge scaling factors were determined by minimizing the unsigned average error between experimental and calculated hydration free energies. The PM3-based charge models do not lead to lower average errors than obtained with the EPS charges for the subset of 13 molecules in the original study. However, improvement is obtained by scaling the CM1A partial charges by 1.14 and the CM3A charges by 1.15, which leads to average errors of 1.0 and 1.1 kcal/mol for the full set of 25 molecules. The scaled CM1A charges also yield the best results for the hydration of amides including the *E*/*Z* free-energy difference for *N*-methylacetamide in water.

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Key words: free energy of hydration; partial atomic charges; charge model; AM1; PM3

Introduction

Molecular mechanics force fields describe the intramolecular and intermolecular potential energy of a molecular system.[1–8] The intermolecular and long-range intramolecular energetics normally consist of Lennard–Jones and Coulomb interactions between atom pairs. Although there is substantial interest in polarizable force fields,[9–11] nonpolarizable models, in which the partial charges on the atomic sites are fixed, still receive the greatest use. Consequently, much effort has gone into the development and testing of alternative procedures for obtaining optimal fixed charges for nonpolarizable force fields. The era of combinatorial chemistry has emphasized the need to have rapid, general procedures that can yield the desired atomic charges for any organic molecule. The dominant route has then been to perform a quantum mechanical calculation and to fit the charges to reproduce electronic properties, especially dipole moments and electrostatic potentials.[12] This has led to several popular alternatives such as CHELPG[13] and RESP[14] charges, which are usually derived from *ab initio* wave functions and faster approaches using, for example, AM1 charges with bond charge corrections.[15]

In 1995, Storer et al. reported the particularly attractive CM1A and CM1P approaches that provide high-quality partial charges from rapid, semiempirical AM1 and PM3 quantum calculations.[16] Their mapping procedures were optimized to yield highly accurate dipole moments for organic molecules. Recent refinements have led to the introduction of Charge Model 3 (CM3), which encompasses a larger training set in conjunction with both semiempirical (AM1 and PM3) and density functional (BLYP and B3LYP) methods.[17] The most significant improvements with CM3 are for the charge distributions and dipole moments of compounds containing N and O, especially amides, nitro compounds, and carboxylic acid derivatives. The correct description of the electrostatics for these molecules has been a long-standing challenge for molecular mechanics force fields. Amines and amides have been particularly problematic including correct reproduction of trends in free energies of hydration (ΔG_{hyd}) with increasing

Correspondence to: W. L. Jorgensen; e-mail: William.Jorgensen@yale.edu

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Table 1. Scaled CM1P and CM3P Charges for the Original 13 Molecules.

Molecule	Atom	1.3*CM1P	1.12*CM3P	Molecule	Atom	1.3*CM1P	1.12*CM3P
C_2H_6	C	−0.137	−0.183	CH_3COOCH_3	C2	−0.158	−0.203
	H	0.046	0.062		H2	0.089	0.099
CH_3Cl	C	0.040	−0.048		C1	0.642	0.516
	H	0.066	0.080		O	−0.616	−0.514
	Cl	−0.239	−0.191		OR	−0.434	−0.338
CH_3OCH_3	C	0.127	0.026		C	0.141	0.037
	H	0.034	0.052		H	0.052	0.068
	O	−0.456	−0.362	CH_3COOH	C2	−0.157	−0.200
CH_3SH	C	−0.138	−0.196		H2	0.090	0.100
	HC	0.090	0.101		C1	0.651	0.524
	S	−0.241	−0.239		O	−0.616	−0.513
	H	0.108	0.131		OH	−0.651	−0.515
CH_3OH	C	0.152	0.047		HO	0.504	0.405
	HC	0.025	0.045	CH_3CONH_2	C2	−0.185	−0.222
	O	−0.695	−0.553		H2	0.080	0.092
	H	0.468	0.371		C1	0.501	0.435
CH_3CN	C2	0.066	−0.001		O	−0.640	−0.571
	H	0.082	0.093		N	−0.793	−0.606
	C1	0.129	0.064		H	0.438	0.343
	N	−0.442	−0.343	C_6H_6	C	−0.132	−0.137
CH_3NH_2	C	−0.020	−0.061		H	0.132	0.137
	HC	0.044	0.062	C_5H_5N	N	−0.447	−0.429
	N	−0.870	−0.705		C1	0.086	0.069
	H	0.379	0.290		C2	−0.195	−0.190
$(CH_3)_2CO$	C1	−0.208	−0.246		C3	−0.071	−0.081
	H	0.083	0.094		H1	0.149	0.151
	C2	0.453	0.373		H2	0.151	0.153
	O	−0.534	−0.446		H3	0.138	0.142

substitution and the differential hydration of *E* and *Z* secondary amides.[18–24]

In previous testing of charge models, we computed free energies of hydration for a set of 13 neutral organic molecules in Monte Carlo simulations using free energy perturbation (MC/FEP) calculations and the TIP4P water model. Among the considered alternatives, the best and similar results were obtained with 6-31G* CHELPG and 1.2*CM1A (CM1A scaled by 1.2) charges.[25,26] Dipole moments are overestimated at the 6-31G* level by 10–20% and a similar enhancement of the CM charges is needed to reflect solute polarization in water.[25,26] The present study was initiated in anticipation of improved results with the CM3 models. Thus, the testing has been expanded by including the CM3A and CM3P models and also by increasing the dataset to 25 molecules including secondary and tertiary amides, a nitro compound, and substituted benzenes. Optimal scaling factors that minimize the errors for computed free energies of hydration were determined through charge perturbations.

Computational Details

Atomic Charges

CM1 and CM3 atomic charges[16,17] were computed via single-point AM1 and PM3 calculations using OPLS-AA[5] optimized geometries and the BOSS program.[27] The program automatically makes charges for many equivalent atoms identical by averaging; this includes hydrogens or halogens in any XY_n group, hydrogens in substituted benzenes, and oxygens in XO_n groups. This is consistent with normal practice for force fields; it avoids potential nonphysical asymmetries, for example, if charges on hydrogens in an XH_n group are not the same, the energy of the system can be different for identical rotamers. It was also shown previously that the averaging has little effect on computed ΔG_{hyd} values, for example, 0.09 ± 0.01 kcal/mol for the NH_2 group and 0.25 ± 0.01 kcal/mol for both the NH_2 and CH_3 groups in acetamide.[25] Computed atomic charges for the molecules in this study are summarized in Tables 1 and 2, and the dipole moments computed from these charges using the OPLS-AA geometries are provided in Tables 3 and 4. As discussed below, the PM3-based charge models are less successful for the ΔG_{hyd} calculations, and results from them are only reported for the original 13 molecules.

Monte Carlo Simulations

Free energies of hydration were computed using the BOSS program[27] and the Monte Carlo/Free Energy Perturbation (MC/FEP) protocol, which has been described in detail.[24–26,30,31] The relative free energy of hydration ($\Delta\Delta G_{hyd}$) between two solutes, A and B, was computed using statistical perturbation

Table 2. Scaled CM1A and CM3A Charges for All 25 Molecules.

Molecule	Atom	1.14*CM1A	1.15*CM3A
C_2H_6	C	−0.237	−0.217
	H	0.079	0.072
CH_3Cl	C	−0.146	−0.162
	H	0.110	0.101
	C1	−0.184	−0.140
CH_3OCH_3	C	−0.052	−0.029
	H	0.079	0.070
	O	−0.372	−0.360
CH_3SH	C	−0.229	−0.186
	HC	0.114	0.104
	S	−0.254	−0.250
	H	0.143	0.123
CH_3OH	C	−0.044	−0.019
	HC	0.181	0.066
	O	−0.587	−0.574
	H	0.407	0.396
CH_3CN	C2	−0.137	−0.103
	H	0.124	0.116
	C1	0.122	0.133
	N	−0.357	−0.377
CH_3NH_2	C	−0.001	−0.013
	HC	0.076	0.068
	N	−1.004	−0.943
	H	0.388	0.376
$(CH_3)_2NH$	C	−0.025	−0.016
	HC	0.078	0.069
	N	−0.802	−0.774
	H	0.382	0.391
$(CH_3)_2CO$	C1	−0.301	−0.272
	H	0.114	0.105
	C2	0.312	0.349
	O	−0.396	−0.436
CH_3COOCH_3	C2	−0.250	−0.220
	H2	0.125	0.115
	C1	0.439	0.455
	O	−0.463	−0.493
	OR	−0.354	−0.335
	C	−0.046	−0.025
	H	0.100	0.091
CH_3COOH	C2	−0.252	−0.220
	H2	0.126	0.117
	C1	0.438	0.456
	O	−0.461	−0.491
	OH	−0.544	−0.528
	HO	0.440	0.432
CH_3CONH_2	C2	−0.280	−0.252
	H2	0.114	0.105
	C1	0.631	0.544
	O	−0.462	−0.568
	N	−1.290	−0.989
	H	0.53	0.475
Z—$CH_3CONHCH_3$	C2	−0.278	−0.251
	H2	0.114	0.104
	C1	0.605	0.544
	O	−0.459	−0.560
	N	−1.096	−0.829
	HN	0.505	0.468
	C	0.099	0.061
	H	0.094	0.084
$CH_3CON(CH_3)_2$	C2	−0.268	−0.240
	H2	0.113	0.104
	C1	0.587	0.550
	O	−0.461	−0.556
	N	−0.904	−0.671
	C	0.075	0.056
	H	0.095	0.082
C_6H_6	C	−0.148	−0.139
	H	0.148	0.139
C_5H_5N	N	−0.352	−0.485
	C1	0.013	0.083
	C2	−0.207	−0.197
	C3	−0.099	−0.086
	H1	0.180	0.172
	H2	0.162	0.154
	H3	0.156	0.147
C_6H_5Cl	C1	−0.034	−0.052
	C2	−0.137	−0.129
	C3	−0.137	−0.129
	C4	−0.147	−0.138
	H2	0.164	0.155
	H3	0.157	0.148
	H4	0.154	0.145
	C1	−0.064	−0.045
C_6H_5OH	C1	0.146	0.137
	O	−0.511	−0.490
	HO	0.433	0.424
	C2	−0.218	−0.211
	C3	−0.104	−0.095
	C4	−0.192	−0.184
	H2	0.157	0.148
	H3	0.152	0.143
	H4	0.151	0.142
$C_6H_5NH_2$	C1	0.208	0.183
	N	−0.909	−0.899
	H	0.408	0.416
	C2	−0.227	−0.218
	C3	−0.100	−0.091
	C4	−0.201	−0.193
	H2	0.149	0.141
	H3	0.147	0.138
	H4	0.147	0.138
$C_6H_5OCH_3$	C1	0.133	0.123
	O	−0.308	−0.285
	C2	−0.205	−0.199
	C3	−0.110	−0.101
	C4	−0.188	−0.180
	H2	0.157	0.148
	H3	0.152	0.143
	H4	0.150	0.141
	C	−0.047	−0.025
	H	0.091	0.082
$C_6H_5CH_3$	C1	−0.082	−0.083
	C2	−0.147	−0.138
	C3	−0.145	−0.136
	C4	−0.154	−0.145
	H2	0.148	0.140
	H3	0.148	0.139
	H4	0.147	0.138

(*continued*)

Table 2. *(Continued)*

Molecule	Atom	1.14*CM1A	1.15*CM3A	Molecule	Atom	1.14*CM1A	1.15*CM3A
E—$CH_3CONHCH_3$	C2	−0.271	−0.242		C	−0.197	−0.168
	H2	0.113	0.104		H	0.092	0.083
	C1	0.610	0.548	$C_6H_5CF_3$	C1	−0.231	−0.230
	O	−0.464	−0.564		C2	−0.09	−0.081
	N	−1.100	−0.831		C3	−0.154	−0.145
	HN	0.512	0.474		C4	−0.116	−0.107
	C	0.105	0.064		H2	0.165	0.156
	H	0.090	0.080		H3	0.160	0.151
$CH_3CH_2NO_2$	C2	−0.250	−0.221		H4	0.156	0.147
	H2	0.115	0.107		C	0.518	0.598
	C1	−0.151	−0.039		F	−0.163	−0.190
	H1	0.150	0.141				
	N	0.682	0.032				
	O	−0.463	−0.187				

theory[32] by converting A to B through a series of intermediate states in the gas phase and in aqueous solution. Mutations involving different number of atoms for A and B were performed using dummy atoms that have no non-bonded parameters (Lennard–Jones and Coulomb).[31] Equilibrium bond lengths for dummy atoms were taken as 0.3 Å, while the bond stretching and angle bending force constants involving dummy atoms are typically set to the corresponding values of their real counterparts.[33] Differences in free energies of hydration were obtained using the thermodynamic cycle shown in Scheme 1, where ΔG_{gas} and ΔG_{aq} were computed using statistical perturbation theory.[30–32] Absolute free energies of hydration are then obtained by connecting all molecules to OPLS methane, whose absolute ΔG_{hyd} has been determined by annihilation.[25,34]

All FEP calculations used 10 windows with double-wide sampling[31] and a coupling parameter λ that ranged from 0 (A) to 1 (B), yielding 20 free-energy increments for each mutation. All internal degrees of freedom were sampled in the MC simulations. The gas-phase MC/FEP calculations consisted of an equilibration phase of 8 million (M) configurations followed by averaging over an additional 8 M configurations. The aqueous systems were generated from an equilibrated box of 500 TIP4P water molecules[35] plus the solute, and the MC simulations were performed in the isothermal-isobaric (NPT) ensemble at 25°C and 1 atm. The cutoffs for solute–solvent and solvent–solvent interactions were set to 10 Å based roughly on center-of-mass separations. The ranges for intramolecular motions and total rotations and translations were adjusted to give overall acceptance rates of ca. 40%. The volume changes have a similar acceptance rate; they are attempted every 3125 configurations in a range of ±250 Å^3. There is no issue of convergence of the volume since it is dominated by the 500 water molecules. Each window consisted of 10 M configurations of equilibration followed by 20–40 M configurations of averaging.

Convergence of the FEP calculations was well established by results for degenerate perturbations, as shown in Table 5. Ideally, the sum of the computed $\Delta\Delta G_{\mathrm{hyd}}$ values should be zero for these cycles. Each cycle involves three FEP calculations; the typical

Table 3. Scaled CM1P and CM3P Dipole Moments (D).

Molecule	μ (EPS)[a]	μ (1.3*CM1P)[b]	μ (1.12*CM3P)[b]	μ (exp)[c]
CH_3OH	2.19	2.24	1.85	1.70
CH_3NH_2	1.79	1.81	1.38	1.31
CH_3CN	4.06	5.09	4.36	3.92
CH_3OCH_3	1.81	1.93	1.68	1.30
CH_3SH	1.99	1.72	1.63	1.52
CH_3Cl	2.25	2.39	2.04	1.89
CH_3CONH_2	4.49	5.68	5.00	3.76
CH_3COOH	1.76	2.42	2.22	1.70
$(CH_3)_2CO$	3.41	3.92	3.44	2.88
CH_3COOCH_3	1.83	2.82	2.48	1.72
Pyridine	2.37	2.20	2.20	2.22

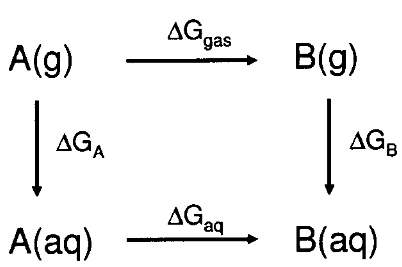
[a]Ref. 25.
[b]Values computed using OPLS-AA optimized geometries.
[c]Ref. 28.

Table 4. Scaled CM1A and CM3A Dipole Moments (D).

Molecule	μ (1.14*CM1A)[a]	μ (1.15*CM3A)[a]	μ (exp)[b]
CH_3OH	2.03	1.96	1.70
CH_3NH_2	1.95	1.82	1.31
$(CH_3)_2NH$	1.81	1.72	1.01
CH_3CN	4.30	4.42	3.92
CH_3OCH_3	1.88	1.77	1.30
CH_3SH	1.75	1.77	1.52
CH_3Cl	2.14	1.90	1.89
CH_3CONH_2	3.73	4.91	3.76
Z—$CH_3CONHCH_3$	3.73	4.96	3.8[c]
E—$CH_3CONHCH_3$	4.30	5.04	
$CH_3CON(CH_3)_2$	4.17	5.01	3.82
CH_3COOH	2.04	2.25	1.70
$(CH_3)_2CO$	3.29	3.48	2.88
CH_3COOCH_3	2.23	2.47	1.72
Pyridine	1.75	2.22	2.22
$CH_3CH_2NO_2$	5.25	4.43	3.23
C_6H_5Cl	1.95	1.74	1.69
C_6H_5OH	1.86	1.79	1.22
$C_6H_5OCH_3$	1.79	1.68	1.38
$C_6H_5CH_3$	0.35	0.35	0.38
$C_6H_5CF_3$	2.82	2.98	2.86
$C_6H_5NH_2$	1.75	1.86	1.13

[a]Values computed using OPLS-AA optimized geometries.
[b]Ref. 28.
[c]Ref. 29.

hysteresis is near 0.3 kcal/mol. These results, the results for batch-means uncertainties for individual FEP calculations, and independent repetitions of some of the absolute ΔG_{hyd} calculations indicate that the statistical uncertainties in the present, computed *absolute* free energies of hydration are no more than 0.5 kcal/mol. When multiple values are available from alternative perturbation cycles, the reported ΔG_{hyd} is for the pathway that involved the smallest structural changes. The new results for free energies of hydration using the CM charges are listed along with experimental and other computed values in Tables 6 and 7; the statistical uncertainties reported for the computed results are $\pm 1\sigma$ as determined by the batch means procedure with batch sizes of 1 M configurations. These uncertainties are consistent with the limit of 0.5 kcal/mol from above. Additional results using the OPLS-AA force field are given for 27 molecules in Table 8; the OPLS-AA parameters including partial atomic charges have been previously reported[5–7] except for recently reoptimized parameters for tertiary amides.[38]



$$\Delta\Delta G_{\text{hyd}}\ (A \rightarrow B) = \Delta G_{\text{aq}} - \Delta G_{\text{gas}} = \Delta G_B - \Delta G_A$$

Scheme 1

Charge Perturbations

The CM charges are scaled linearly for neutral molecules to reflect polarization of the solutes in condensed phases. Preliminary scaling factors, α, were obtained by correlating the CM charges for 210 small organic molecules to CHELPG charges[13] from *ab initio* 6-31G* wave functions. These initial α values were 1.1, 1.3, 1.2, and 1.0 for CM1A, CM1P, CM3A, and CM3P, respectively. The first round of MC/FEP calculations was performed with these scaling factors to yield absolute free energies of hydration. The ΔG_{hyd} for a new α-value is then just the initial value plus the ΔG for the charge change, $\Delta\alpha$, in water minus the ΔG for the charge change in the gas phase. The gas-phase change is not zero because the intramolecular nonbonded energy is altered. Optimal α-values were determined by perturbing from the initial values in increments $\Delta\alpha = 0.01$ until the errors in the computed free energies of hydration were minimized. The FEP results for the charge perturbations converge readily; 8 M configurations of equilibration and averaging in the gas phase, and 5 M configurations of equilibration and 10 M configurations of averaging in aqueous solution were more than adequate.

Table 5. Computed Net Changes in Free Energies of Hydration for Closed Cycles.

	Hysteresis (kcal/mol)	
Conversion	1.14*CM1A	1.15*CM3A
$PhNH_2 \rightarrow PhOH \rightarrow PhH \rightarrow PhNH_2$	0.21	−0.10
$PhOCH_3 \rightarrow PhOH \rightarrow PhH \rightarrow PhOCH_3$	0.48	0.51
$CH_3OH \rightarrow CH_3SH \rightarrow CH_4 \rightarrow CH_3OH$	−0.62	−0.34
$AcN(CH_3)_2 \rightarrow E\text{-}AcNHCH_3 \rightarrow AcNH_2 \rightarrow AcN(CH_3)_2$	−0.14	−0.30
$E\text{—}AcNHCH_3 \rightarrow AcOH \rightarrow AcNH_2 \rightarrow E\text{-}AcNHCH_3$	−0.21	0.12
$AcNH_2 \rightarrow Z\text{-}AcNHCH_3 \rightarrow AcOH \rightarrow AcNH_2$	−0.69	0.39
$Z\text{—}AcNHCH_3 \rightarrow AcOH \rightarrow E\text{-}AcNHCH_3 \rightarrow Z\text{-}AcNHCH_3$	−0.56	0.76
$CH_3OH \rightarrow CH_4 \rightarrow CH_3NH_2 \rightarrow CH_3OH$	−0.36	0.45
$CH_3OH \rightarrow CH_4 \rightarrow CH_3CH_3 \rightarrow CH_3OH$	0.30	−0.33
$CH_3OH \rightarrow CH_4 \rightarrow CH_3OCH_3 \rightarrow CH_3OH$	0.36	0.50

E *to* Z *Conversion for NMA*

To examine the difference in free energy of hydration between *E* and *Z* *N*-methylacetamide (NMA), analogous FEP calculations were used to interconvert the conformers by perturbing the OCNH dihedral angle. An increment, $\Delta\phi$, of 9° was used, resulting in 10 windows with double-wide sampling to cover the 180° range. The gas-phase FEP calculations consisted of 8 M configurations of equilibration followed by 8 M configurations of averaging, while the corresponding lengths were 10 M and 40 M for the FEP calculations in water. The FEP calculations were performed using OPLS-AA, 1.14*CM1A, and 1.15*CM3A charge distributions for the *Z* isomer; the differences in CM charges for the *E* and *Z* forms are slight (Table 2).

Results and Discussion

Absolute Free Energies of Hydration

To compute an absolute free energy of hydration, which corresponds to transfer of a molecule from the gas phase into aqueous solution, the molecule needs to be annihilated in both media.[34] However, it is sufficient to do this for just one anchor molecule and then compute relative free energies of hydration through perturbations to it. The requisite annihilations were performed for OPLS united-atom methane in water and for all-atom methane both in the gas phase and in water. For all-atom methane, two simulations were performed— one to neutralize the partial charges, and the other for the Lennard–Jones contributions. Although the current

Table 6. Calculated and Experimental Free Energies of Hydration (kcal/mol) for the Original 13 Molecules.

Molecule	ΔG_{hyd}, EPS[a]	ΔG_{hyd}, 1.2*CM1A[b]	ΔG_{hyd}, 1.3*CM1P	ΔG_{hyd}, 1.12*CM3P	ΔG_{hyd}, exp[c]
CH_3OH	−4.6 ± 0.4	−4.4 ± 0.4	−4.9 ± 0.1	−2.3 ± 0.2	−5.10
CH_3NH_2	−4.3 ± 0.5	−6.5 ± 0.5	−2.8 ± 0.2	−0.5 ± 0.1	−4.56
CH_3CN	−4.7 ± 0.5	−5.0 ± 0.5	−6.2 ± 0.1	−3.4 ± 0.1	−3.89
CH_3OCH_3	−1.4 ± 0.5	−2.6 ± 0.5	−1.3 ± 0.2	−0.4 ± 0.1	−1.91
CH_3SH	0.0 ± 0.5	0.7 ± 0.5	1.4 ± 0.2	1.6 ± 0.1	−1.24
CH_3Cl	0.1 ± 0.5	0.1 ± 0.5	0.6 ± 0.2	0.9 ± 0.1	−0.55
C_2H_6	2.7 ± 0.4	2.5 ± 0.5	3.0 ± 0.2	3.1 ± 0.1	1.83
CH_3CONH_2	−13.4 ± 0.4	−11.9 ± 0.5	−16.1 ± 0.3	−13.5 ± 0.2	−9.71[d]
CH_3COOH	−8.5 ± 0.4	−7.7 ± 0.4	−7.7 ± 0.2	−6.4 ± 0.2	−6.70
$(CH_3)_2CO$	−5.4 ± 0.5	−3.6 ± 0.5	−4.6 ± 0.3	−3.9 ± 0.2	−3.81
CH_3COOCH_3	−5.3 ± 0.5	−4.6 ± 0.5	−4.5 ± 0.3	−4.9 ± 0.2	−3.32
C_6H_6	−0.4 ± 0.4	−2.9 ± 0.4	−0.3 ± 0.2	−0.8 ± 0.2	−0.86
Pyridine	−4.9 ± 0.4	−5.1 ± 0.4	−3.4 ± 0.2	−3.4 ± 0.2	−4.70
Avg. error	1.2	1.1	1.6	1.3	

[a]Ref. 25.
[b]Ref. 26.
[c]Ref. 36.
[d]Ref. 37.

Table 7. Calculated and Experimental Free Energies of Hydration (kcal/mol) Using Scaled CM1A and CM3A Charges.[a]

Molecule	1.14*CM1A	1.15*CM3A	Expt[a]
C_2H_6	2.55 ± 0.11	2.98 ± 0.16	1.83
C_3H_8	3.07 ± 0.12	3.15 ± 0.17	1.96
CH_3Cl	0.72 ± 0.11	1.28 ± 0.11	−0.55
CH_3OCH_3	−1.26 ± 0.15	−0.33 ± 0.24	−1.91
CH_3SH	1.24 ± 0.11	1.23 ± 0.21	−1.24
CH_3OH	−2.67 ± 0.15	−2.56 ± 0.24	−5.10
CH_3NH_2	−5.48 ± 0.14	−3.91 ± 0.13	−4.56
$(CH_3)_2NH$	−3.34 ± 0.17	−3.12 ± 0.15	−4.30
$(CH_3)_2CO$	−2.96 ± 0.16	−3.45 ± 0.20	−3.81
CH_3COOCH_3	−4.35 ± 0.20	−4.96 ± 0.24	−3.32
CH_3COOH	−6.34 ± 0.17	−6.61 ± 0.21	−6.70
CH_3CONH_2	−10.24 ± 0.19	−13.32 ± 0.23	−9.71[b]
Z—$CH_3CONHCH_3$	−9.48 ± 0.23	−12.60 ± 0.26	−10.08[b]
E—$CH_3CONHCH_3$	−9.13 ± 0.21	−9.66 ± 0.27	−10.08[b]
$CH_3CON(CH_3)_2$	−9.30 ± 0.28	−8.34 ± 0.30	−8.55[b]
CH_3CN	−3.21 ± 0.14	−3.54 ± 0.15	−3.89
$CH_3CH_2NO_2$	−6.37 ± 0.14	−3.15 ± 0.17	−3.71
C_6H_6	−1.41 ± 0.23	−1.23 ± 0.24	−0.86
Pyridine	−3.15 ± 0.24	−5.10 ± 0.25	−4.70
C_6H_5Cl	−0.93 ± 0.24	−0.72 ± 0.25	−1.12
C_6H_5OH	−5.42 ± 0.24	−5.48 ± 0.25	−6.62
$C_6H_5OCH_3$	−2.70 ± 0.26	−2.30 ± 0.26	−2.46
$C_6H_5CH_3$	−1.58 ± 0.24	−1.24 ± 0.25	−0.89
$C_6H_5CF_3$	−0.51 ± 0.24	−0.40 ± 0.25	−0.25
$C_6H_5NH_2$	−7.57 ± 0.25	−8.39 ± 0.38	−5.49
Avg error	1.03	1.13	

[a]Ref. 36.
[b]Ref. 37.

simulations used twice as many water molecules as before, the present ΔG_{hyd} for united-atom methane of 2.47 ± 0.1 kcal/mol agrees with the previously computed values of 2.47 ± 0.36[25] and 2.3 ± 0.3[34] kcal/mol. The experimental value is 2.005 kcal/mol.[39] The present result for OPLS all-atom methane is 2.56 ± 0.10 kcal/mol; prior values have been reported using rigid geometries, 2.20 ± 0.22 kcal/mol with 265 TIP4P water molecules[40] and 2.40 ± 0.04 with 512 TIP4P water molecules.[41] The first seven molecules in Table 6 were converted to methanol, which was converted to all-atom methane. Acetic acid derivatives were converted to acetic acid, which was converted to acetone and subsequently to propane and methane. ΔG_{hyd} results for benzene derivatives were obtained by conversion to benzene and then to all-atom methane; in an earlier study, rigid all-atom benzene was perturbed to united-atom methane.[42] To be clear, the typical FEP step here has an uncertainty of less than ca. ±0.1 kcal/mol, and the series of steps needed to compute an absolute free energy of hydration yields uncertainties of less than ca. ±0.5 kcal/mol.

CM1P and CM3P Charge Models

As shown in Table 6, the average unsigned errors for the original 13 molecules are 1.2 and 1.1 kcal/mol from the previous reports with the 6-31G* and 1.2*CM1A charges.[25,26] The PM3-based methods were tested for the original 13 molecules, as summarized in Table 6. ΔG_{hyd} values were obtained with 1.3*CM1P and 1.0*CM3P charges. The average unsigned error with 1.3*CM1P of 1.6 kcal/mol, the large error for acetamide, and the pattern of errors indicated that optimization of the charge scaling factor would not lead to a competitive procedure, so further CM1P computations were not performed. The CM3P results were more promising, and the optimal scaling factor was determined to be 1.12 following the protocol described above. Nevertheless, the results in Table 6 with an average error of 1.3 kcal/mol for 1.12*CM3P did not show improvement over the prior results with the 6-31G* EPS or 1.2*CM1A charges. With 1.12*CM3P, the saturated compounds are consistently undersolvated, including a 4 kcal/mol error for methylamine, while the error for acetamide is 4 kcal/mol in the other direction. From a bioorganic perspective, amines and amides are critically important, so these errors are not acceptable. It is possible that the PM3-based charges could be made viable by modifying the OPLS-AA Lennard–Jones (LJ) parameters. This complication was not pursued. An intended use for the CM charges is in QM/MM calculations for which it would be inconvenient to use different LJ parameters in the QM and MM parts. Other alternatives could also be pursued such as obtaining the CM

Table 8. Calculated and Experimental Free Energies of Hydration (kcal/mol).

Molecule	OPLS-AA[a]	Expt[b]
CH_4	2.56	2.01
C_2H_6	2.21	1.83
C_3H_8	2.43	1.96
n-C_4H_{10}	2.72	2.08
c-C_6H_{12}	0.87	1.23
CH_3OCH_3	−1.83	−1.91
CH_3SH	−1.92	−1.24
CH_3OH	−5.16	−5.10
C_2H_5OH	−5.07	−5.01
n-C_3H_7OH	−4.41	−4.86
CH_3NH_2	−3.87	−4.56
n-$C_3H_7NH_2$	−3.37	−4.39
$(CH_3)_2NH$	−3.77	−4.30
$(CH_3)_3N$	−2.24	−3.21
$CH_3COCH_2CH_3$	−2.94	−3.71
CH_3CH_2COOH	−5.26	−6.47
CH_3CONH_2	−9.65	−9.71[c]
Z-$CH_3CONHCH_3$	−7.86	−10.08[c]
E-$CH_3CONHCH_3$	−7.47	−10.08[c]
$CH_3CON(CH_3)_2$	−8.45	−8.55[c]
CH_3CN	−4.14	−3.89
$CH_3CH_2NO_2$	−3.01	−3.71
C_6H_6	−0.35	−0.86
$C_6H_5CH_3$	−0.46	−0.89
$C_6H_5CF_3$	0.56	−0.25
C_6H_5Cl	0.15	−1.12
$C_6H_5NH_2$	−4.78	−5.49
Avg error	0.69	

[a]Uncertainties are ± 0.3–0.5 kcal/mol.
[b]Ref. 36.
[c]Ref. 37.

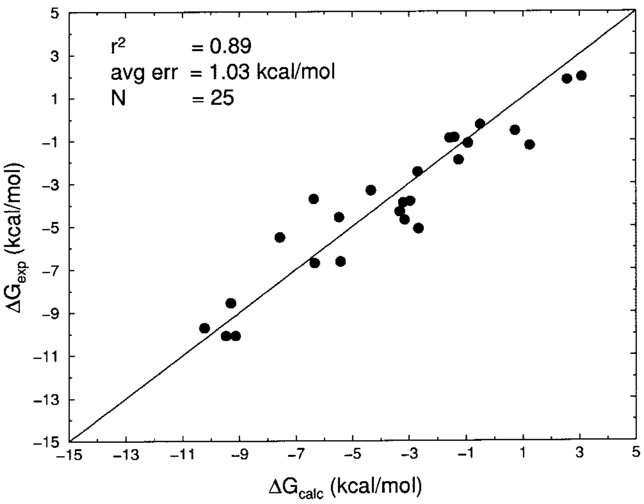


Figure 1. Correlation of experimental and computed absolute free energies of hydration using 1.14*CM1A charges.

charges by including an aqueous reaction field in the AM1 or PM3 calculations[16,17] rather than applying the present constant scaling.

CM1A and CM3A Charge Models

Because the results with the CM1P and CM3P procedures were not satisfactory, the CM1A and CM3A methods were focused upon. Furthermore, the original test set was expanded to 25 molecules including propane, nitroethane, dimethylamine, *E*- and *Z*-NMA, *N*,*N*-dimethylacetamide (DMA), and six benzene derivatives. The additions were chosen to increase the diversity of the dataset and to be particularly challenging. Optimal scaling factors of 1.14 and 1.15 for CM1A and CM3A charges were obtained by minimizing the errors in the computed hydration free energies. The optimized scaling factor of 1.14 for CM1A is close to the previously determined α of 1.2 obtained using the smaller dataset.[26]

With average unsigned errors of 1.03 and 1.13 kcal/mol (Table 7), the overall performance of the 1.14*CM1A charge model is slightly superior to that of 1.15*CM3A. The mean signed errors for the 1.14*CM1A and 1.15*CM3A results on the 25 molecules are +0.26 and +0.15 kcal/mol, which somewhat reflects the overestimate of the free energy of hydration of OPLS-AA methane by 0.55 kcal/mol. The error with the 1.15*CM3A charges has a significant contribution from overly exoergic values of ΔG_{hyd} for acetamide and *Z*-NMA, while the largest error (2.7 kcal/mol) with the 1.14*CM1A charges is for nitroethane. Both procedures also yield errors greater than 2 kcal/mol for methanol and methanethiol, while the performances for phenol are better. The methods do yield the qualitatively correct order for methylamine and dimethylamine, but err for acetamide vs. NMA. It was thought that a trifluoromethyl group might be problematic, but the results for trifluoromethylbenzene are in excellent accord with experiment. In fact, the results for the aromatic compounds are generally good; the computed free energies of hydration for benzene, toluene, trifluoromethylbenzene, chlorobenzene, and anisole are all within 0.7 kcal/mol of the experimental values, using either charge model. The principal outlier for this series is aniline, with errors of 2–3 kcal/mol.

Plots of the experimental vs. computed free energies of hydration are presented in Figures 1 and 2. Good linear relationships are obtained using both charge models with correlation coefficients, r^2, of 0.89 and 0.90 for 1.14*CM1A and 1.15*CM3A, respectively. In view of the importance of amides and the lower average error with the 1.14*CM1A charges, they are recommended for general bioorganic applications. It should be emphasized that the typical experimental uncertainty in measurements of free energies of hydration is ca. 0.3 kcal/mol,[36,37] and the statistical noise in the present MC/FEP results is ca. 0.5 kcal/mol. Thus, average agreement between FEP and experimental values of better than ca. 0.6 kcal/mol should not be achievable.

The present results can be compared with those from the recent work by Jakalian et al.[15] They developed a competitive method using AM1 charges from a population analysis that are adjusted using many bond charge corrections (BCCs) to yield improved electrostatic potentials and hydrogen-bond strengths. Although they did not compute absolute free energies of hydration, they did report 40 FEP results for relative free energies of hydration in TIP3P water for conversions mostly of polar molecules to isosteric hydrocarbons. Previous comparisons show negligible differences for computations of free energies of hydration in TIP3P or TIP4P water.[43] For the 40 $\Delta\Delta G_{\mathrm{hyd}}$ values, the average unsigned errors, which should be lower than for absolute free energies of hydration, are 2.6, 1.4, 1.1, and 0.7 kcal/mol using uncorrected AM1 charges, 6-31G* RESP charges, charges from the MMFF force field, and the AM1-BCC charges. As they note, to improve the AM1–BCC results, five BCCs were specifically modified for amines and nitro compounds, which, for example, lowered the error for the nitroethane to isopentane conversion from 5.0 to 0.5 kcal/mol. In any event, 1.14*CM1A and AM1-BCC charges are certainly preferred to the alternatives including 6-31G* RESP or CHELPG charges. Simple BCC adjustments for the few problem cases, especially nitro compounds, could clearly enhance the 1.14*CM1A results.

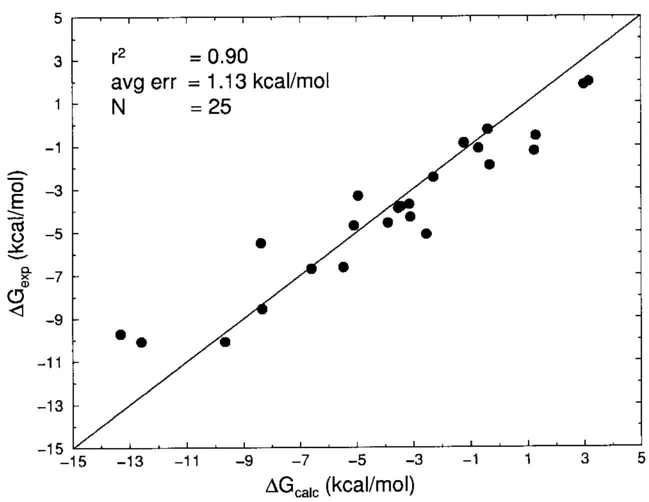


Figure 2. Correlation of experimental and computed absolute free energies of hydration using 1.15*CM3A charges.

Table 9. Computed E–Z Free Energy Differences (kcal/mol) for *N*-Methylacetamide in the Gas Phase and in Water at 25°C.

	1.14*CM1A	1.15*CM3A	OPLS-AA
ΔG_{gas}	1.72	1.28	2.84
ΔG_{aq}	2.00	4.19	3.23
$\Delta\Delta G_{hyd}{}^{a}$	0.28	2.91	0.39
$\Delta\Delta G_{hyd}{}^{b}$	0.35[b]	2.94[b]	

[a]Difference in free energies of hydration for *E*- and *Z*-NMA from the FEP calculations driving the OCNH dihedral angle; experimental value is near zero (ref. 37b).
[b]From the computed ΔG_{hyd} values in Table 7.

However, key advantages of staying with the unmodified 1.14*CM1A charges are the avoidance of having to perceive or add bond types and the inapplicability of BCC methods to studying reaction paths with changes of bond types, as in QM/MM simulations of reactions in solution.[26,44]

OPLS-AA Results

The free energies of hydration using the full OPLS-AA force field are generally in close accord with experiment (Table 8) with the exception of NMA. It is too hydrophobic, and the same pattern for NMA and acetamide has been noted for the AMBER all-atom and OPLS-UA force fields.[20,21] In the original OPLS-AA report for amides,[5] the parameters were validated through reproduction of gas-phase conformational energetics and properties of pure liquid amides; however, free energies of hydration were not computed. The latter calculations became routine, and the benefits can be seen in the results for the amines,[6] acetonitrile,[7] nitroethane,[7] and dimethylacetamide[38] in Table 8. Modified OPLS-AA parameters for secondary amides that remedy the remaining errors in ΔG_{hyd} have been developed, and will be reported elsewhere.[45]

E *and* Z *NMA*

An important observation for secondary amides is that their *E*/*Z* equilibrium shows negligible solvent effects.[37b,46] For NMA, the *Z* form (φ(OCNH) = 180°) is lower in free energy than the *E* conformer by ca. 2.5 kcal/mol in the gas-phase and in aqueous solution.[18,37b,46–48] The similar hydration of the two conformers is reasonably well reproduced by the 1.14*CM1A and OPLS-AA results in Tables 7 and 8; however, the *E* form is erroneously predicted to be less well hydrated than the *Z* form by nearly 3 kcal/mol with the 1.15*CM3A charges. The dipole moments for the *Z* and *E* isomers are 3.73 and 4.30 D with the 1.14*CM1A charges (Table 4), and 4.04 and 4.16 D with OPLS-AA. The corresponding values are larger with the 1.15*CM3A charges, 4.96 and 5.04 D, which contributes to the more negative free energies of hydration, but does not explain the *Z*/*E* difference. The dipole moments are greater for the *E* conformer; however, the hydrogen bonding sites, the carbonyl O and amide H, are closer together in the *E* form, and presumably interfere with formation of optimal hydrogen bonding simultaneously at both sites. The larger difference in dipole moments with the 1.14*CM1A charges appears to help balance the two effects better and leads to the smaller difference in free energies of hydration.

To confirm the differences in ΔG_{hyd} for *Z*- and *E*-NMA and to determine their total free energy differences in the gas phase and in aqueous solution, the additional FEP calculations were performed to perturb the OCNH dihedral angle from 180° to 0°. The results are summarized in Table 9; the statistical uncertainties ($\pm 1\sigma$) for the computed quantities are ca. $\pm$0.05 kcal/mol in the gas phase and $\pm$0.15 kcal/mol in water. With the OPLS-AA potential functions, the computed *E*–*Z* free-energy difference is 2.8 kcal/mol at 25°C and 1 atm in the gas phase. In water, the computed ΔG of 3.2 kcal/mol is a little greater than the experimental value of 2.5 $\pm$ 0.4 kcal/mol,[46] and the *E* conformer is computed to be less well hydrated than the *Z* form by 0.4 kcal/mol. This value was used to yield the ΔG_{hyd} for *E*-NMA in Table 8 from the ΔG_{hyd} for *Z*-NMA. However, the ΔG_{hyd} values for the two conformers with the CM charges were determined independently in Table 7. Change to the CM charges and use of the OPLS-AA parameters otherwise lowers the gas-phase ΔG to 1.7 and 1.3 kcal/mol for 1.14*CM1A and 1.15*CM3A. The dihedral angle driving then yields ΔG values in water of 2.0 and 4.2 kcal/mol, respectively. Thus, the *Z* conformer is computed to be better hydrated than the *E* form by 0.3 and 2.9 kcal/mol using the 1.14*CM1A and 1.15*CM3A charge models. These results agree exactly with the differences obtained from the annihilation protocol for Table 7.

Patterns for the Charge Distributions and ΔG_{hyd}

In attempting to analyze the origin of the errors in the computed free energies of hydration, one can start by plotting the errors for dipole moments for the unscaled CM models vs. the ΔG_{hyd} errors. This reveals only weak correlations with r^2 values of roughly 0.2. Overall, the dipole moment errors are small with the CM methods.[16,17] The only case with errors above 1 D is nitroethane, and this does lead to overly favorable ΔG_{hyd} values especially with the 1.14*CM1A charges (Table 7). Similarly, the lower dipole moments for amides with CM1A than CM1P, CM3A and CM3P translate to improved free energies of hydration. However, for other cases such as methanol and methanethiol, the dipole moment errors are small with the CM models, but the errors for ΔG_{hyd} are relatively large. Thus, details of the charge distributions are critical in determining the quality of the ΔG_{hyd} results.

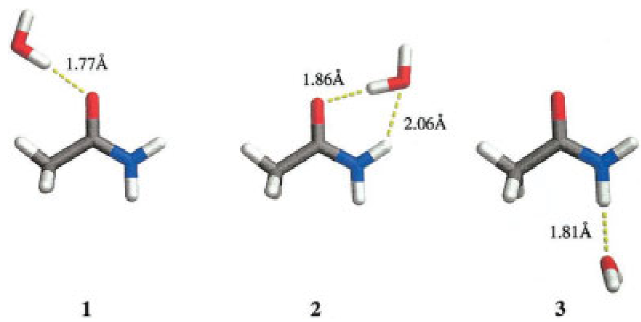


Figure 3. Optimized acetamide–water complexes using the OPLS-AA force field.

Table 10. Computed Interaction Energies (kcal/mol) for the Acetamide–Water Complexes in Figure 3.

Complex	1.14*CM1A	1.15*CM3A	OPLS-AA
1	−6.49	−8.54	−6.92
2	−9.25	−11.44	−9.31
3	−7.18	−8.06	−6.75

For amides, the charges for the OCNH unit are obviously important for controlling hydrogen-bond strengths and ΔG_{hyd}. As an example, the OPLS-UA model of acetamide yields the correct ΔG_{hyd} and the charges are O (−0.50), C (0.50), N (−0.85), and H (0.425).[43] The OPLS-AA model also does well (Table 8), and the charges are O (−0.50), C (0.50), N (−0.76), and H (0.38).[5] By comparison, the 1.3*CM1P and 1.12*CM3P models (Table 1) and the 1.15*CM3A model (Table 2) feature greater polarization of the C=O bond with charges near −0.6 on oxygen. The 1.14*CM1A charge of −0.46 for the oxygen is closer to the OPLS values; however, the nitrogen is the most negative (−1.29) and the hydrogen is the most positive (0.53) among the models. The effects of these differences are reflected in results for the complexes of acetamide and a single TIP4P water molecule that are illustrated in Figure 3. The interaction energies from gas-phase optimizations are listed in Table 10. Results of *ab initio* calculations at the CBS-Q level for the analogous formamide–water complexes are similar to the OPLS-AA results in Table 10 with interaction energies of −6.66, −9.61, and −5.37 kcal/mol for complexes **1–3,** respectively.[49] The OPLS-AA and 1.14*CM1A interaction energies for complexes **1** and **2** with the water molecule hydrogen bonded to the oxygen are similar, while the greater negative charge on oxygen (−0.57) with the 1.15*CM3A model contributes to strengthening of the hydrogen bonds by 2–3 kcal/mol. The interaction energy for complex **3** with the water molecule hydrogen bonded to the *trans* amide hydrogen is also too attractive by ca. 2 kcal/mol with the 1.15*CM3A charges. Clearly, the strengthening of the three hydrogen bonds by 2–3 kcal/mol each can readily translate into the 3–4 kcal/mol overly exoergic ΔG_{hyd} for acetamide in Table 7.

For the 1.14*CM1A model, the most serious problem is with nitroethane. The dipole moment is much too large (Table 4), which stems from the substantial charges on nitrogen (0.68) and oxygen (−0.46) in Table 2. For comparison, the corresponding OPLS-AA charges are 0.54 and −0.37.[7] The opposite problem occurs for methanol, for which the 1.14*CM1A and 1.15*CM3A charges are near −0.58 and 0.40 for the oxygen and hydroxyl hydrogen, while they are −0.683 and 0.418 with OPLS-AA, which yields the correct ΔG_{hyd} (Table 8). The net message is well known in force-field development; namely, hydrogen bonding and, therefore liquid-state properties, are very sensitive to details of molecular charge distributions. In view of this and the underlying simplicity of the charge models with just a single interaction site on each atom, the quality of the computed results in Tables 7 and 8 is actually remarkable.

Conclusion

Partial atomic charges are needed for the description of molecules in nonpolarizable force fields, which continue to be used for many purposes from gas-phase conformational searching to molecular dynamics simulations of proteins and nucleic acids in solution. In view of the diversity of organic molecules, general procedures are needed for the rapid computation of partial atomic charges. The utility of CM1 and CM3 charge models based on AM1 and PM3 wave functions has been examined here for neutral molecules using the important test of computing free energies of hydration. Optimal scaling factors were determined and the most promising charge models that emerged are 1.14*CM1A and 1.15*CM3A. Average unsigned errors of 1.03 and 1.13 kcal/mol were obtained with these models, respectively, for the free energies of hydration of 25 diverse organic molecules using the TIP4P potential for water. Description of the hydration of amides is significantly better with the 1.14*CM1A charges including the *E*/*Z* free-energy difference for *N*-methylacetamide in water, while the greatest flaw is for the difficult case of nitro compounds. Overall, the 1.14*CM1A charge model provides superior and faster performance than many extensively utilized alternatives including 6-31G* CHELPG and RESP charges, and it can be recommended for general use and further study.

Acknowledgments

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